



EFFECTS OF *PILIOSTIGMA THONINGII* (KALGO) LEAVES DIETS ON PERFORMANCE AND DIGESTIBILITY OF KANO BROWN BUCKS IN SEMI-ARID ZONE OF NIGERIA

^{*1}Ashiru, R. M., ²Iliyasu, K. A., ³Madu, A., ⁴Babandi, B., ⁵Ubayi, K. M., ⁶Ado, F. H., ⁷Sariyyu, S., ⁸Afolabi, H.O., ⁹Danzabuwa., A. A. and ¹⁰Jibrin, I.

¹Department of Animal Science, Kano University of Science and Technology, Wudil, Kano State, Nigeria., ²Kano Agricultural and Rural Development

Authority, (KNARDA), Kano State, Nigeria., ³Department of Animal Health and Production Technology, Federal College of Agricultural Produce Technology, Kano State, Nigeria., ⁴Department of Animal Health and Production, Binyaminu Usman Polytechnic, Hadejia, Jigawa State, Nigeria.

⁵Department of Agricultural Education Sa'adatu Rimi University of Education, Kumbotso, Kano State Nigeria, ⁶Department of Animal Production Audu Bako College of Agriculture, Dambatta, Kano State, Nigeria. ⁷Department of Biological Sciences, Bayero University, Kano, Nigeria.

⁸Department of Animal Science, Federal University Dutsinma (FUDMA), Katsina State, Nigeria. *Corresponding Author: ashirurabi@yahoo.co.uk

+234)8037037041

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ABSTRACT

Key words:

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The study was conducted to determine the effects of feeding diet of *Piliostigma thoningii* leaves on performance and digestibility of Kano Brown bucks in semi-arid zone of Nigeria. Sixteen Kano Brown bucks were divided into four treatment groups of four animals per treatment viz; T1 (0%) as control, T2 (15%), T3 (30%) and T4 (45%) inclusion levels of *Piliostigma thoningii* leaves in a completely randomized design (CRD). The animals were fed *ad-libitum*. The feeding trial lasted for 12 weeks (84 days). Parameters measured and calculated include; initial body weight, average weight gain, feed intake, , feed efficiency as well as nutrient Digestibility. The results showed that there were no significant differences ($P > 0.05$) in all the growth performance parameters evaluated. However, significant ($P < 0.05$) differences in digestibility parameters except ash digestibility. It is concluded that *Piliostigma thoningii* leaves could be included in complete diets up to 45% for Kano Brown bucks. It is therefore, recommended that 45% of *Piliostigma thoningii* could be included in complete ration to Kano Brown bucks.

INTRODUCTION

Forage trees and shrubs are alternative fodder resources in ruminant nutrition and have high potential to maximize productivity. Several indigenous browse species are available hence the need for continuous investigation and screening to identify and utilize those with good nutritional potentials (Aruwayo, and Adeleke, 2022). Exploitation of cheap feed resources for animal production would lower the market price of animals and their products, it also improve the intake of animal protein by the general populace in developing countries such as Nigeria (Ayaode *et al.*, 2007). Browse plants form a significant part of ruminant's diet in the semi-arid region especially in the area where meat is a major source of protein. There are several numbers of browse plants that are under-utilized by the livestock farmers. *Piliostigma thoninngii* (Kargo) is regarded as a plant with good fodder value in the Northern part of Nigeria and other parts of the arid and semi-arid areas of tropical and sub-tropical Africa. Traditional herdsman cut the branches of these trees and feed the leaves and pods to livestock during the dry season when other forages are not available (Jokhthan *et al.*, 2010). *Piliostigma thoningii* known locally as 'Kargo' or 'Kalgo' is

one of the medium-small size leguminous shrub that grow in the tropics. It is well-known browse specie in Northern Nigeria (Akin-Osanaiye *et al.*, 2009). The *P. thonningii* fruits have been reported to be eaten by wild animals while Farmers in the lower Savanna region grind the seeds/pods as feed for ruminants during dry season (Jimoh and Oladiji, 2005). Iliyasu *et al.* (2019) reported that *Piliostigma thonningii* leaves contain anti-nutritional factors; 0.03% oxalate, 0.21% phytate, 0.10% tannin and 0.37% saponin. This study therefore, broadly aimed at determining the effects of feeding diets containing varying levels of *Piliostigma thonningii* leaves on growth performance and digestibility of Kano Brown bucks. However, the study was specifically to determine feed intake, weight gain, total feed consumed, feed efficiency and feed conversion ratio, as well as proximate composition of the experimental diets and fecal samples for digestibility determination.

MATERIALS AND METHODS

Study Location

The experiment was conducted at Kano University of Science and Technology, Wudil Commercial and Research Farm Gaya. The farm is located on latitude 11°51'N and longitude 92°E and altitude of 400 – 450 m above sea level. The annual rainfall is 600 mm per annum with relative humidity of 75% during the raining season occurring from May to October with peak in August, the region is therefore characterized by long dry season of about 7 months. The mean annual temperature is 26°C (Olofin, 2008).

Experimental animals and management

Sixteen (16) Kano Brown bucks (averagely weighing 13.75±0.25; aged (8 – 10 month) were sourced from the University Farm and quarantined for two weeks. The bucks were treated for internal and external parasites with Ivomec® – super at 10mg/kg body weight subcutaneously before the commencement of the experiment. The experimental animals were managed intensively in the farm and group fed before commencement of the experiment. The sixteen (16) experimental animals were divided into four groups with four animals per treatment. They were blocked based on live weight to balance each treatment. They were house in individual pens measuring 2 by 1 m. The pens were cleaned and disinfected using detergent and Moriguard. Water and salt lick were provided *ad-libitum*.

Experimental Feed Preparation and Formulation

Four (4) complete diets were formulated containing *Piliostigma thonningii* leaves at graded levels of 0, 15, 30 and 45%. The diets were offered *ad-libitum* as complete diet to all the experimental animals.

Table 1: Composition (%) of Experimental Diets

Ingredients	Treatments			
	I(0%)	II(15%)	III(30%)	IV(45%)
<i>P. thonningii</i> leaves	0	15	30	45
Cowpea husk	20	20	20	20
Wheat bran	20	20	15	15
Rice milling waste	15	10	10	0
Groundnut haulms	20	15	15	10
Maize	19	14	9	9
Soybean	5	5	5	5
Salt	1	1	1	1
TOTAL	100	100	100	100

Experimental Design

The experimental animals were randomly allotted to four (4) treatments with four (4) replications per treatment in a completely randomized design (CRD).

Parameters measured**Feed Intake and Live Weight**

Daily records of feed intake were taken throughout the period of the experiment by weighing feed offered minus left over. The animals were weighed prior to commencement of the experiment and subsequently at weekly intervals between 08:00 and 09:00 am before offering the feed. The feeding trial lasted for 12 weeks (84 days). Other growth parameters calculated include; average daily gain (ADG); was obtained by dividing total weight gain by the number of days of the experiment, feed conversion ratio (FCR); was calculated as feed intake divided by weight gain, feed efficiency (FE); was calculated as weight gain divided by total feed intake multiply by 100.

Laboratory Analysis

Thoroughly mixed representative samples of the diets and faecal samples were analyzed for proximate composition (dry matter, ether extract, nitrogen free extract, crude protein, crude fibre, ash) according to AOAC (2005) procedures.

Statistical Analysis

The collected data were analyzed using analysis of variance (ANOVA) procedures of the Statistical Analytical Systems (SAS, 2009) software package version 9.2 and the means with significant differences were separated using Least Significant Difference (LSD) at 5% level of probability ($P < 0.05$).

RESULTS AND DISCUSSION

Proximate Composition of Experimental Diets and Test Material

The results for proximate composition of test material and experiment diets were presented in Table 2. The values of dry matter, crude protein, crude fibre, ether extract, Ash and nitrogen free extract obtained in this study were 95.56%, 11.61%, 25.23%, 3.09% 7.43% and 52.64 respectively for test material (PTL). The values of DM, CP and CF obtained in this study were higher than values (92.50%, 11.49%, and 21.48% respectively) reported by Iliyasu *et al.* (2019) for *P. thoningii* leaves in semi-arid region in Nigeria. However, the values of EE, Ash and NFE obtained in this study were lower than values (5.75%, 9.55% and 56.74% respectively reported by Iliyasu *et al.* (2019) for *P. thoningii* leaves in semi-arid region in Nigeria. The variations possibly occurred due to maturity stage and time when sample of test material was collected. However, the values of proximate composition of experimental diets, dry matter vary from 94.30 - 94.88% with T₄ having the numerically higher value. Crude protein values ranged from 12.17 - 13.63% with T₂ having numerically the higher value. The crude fibre content varies from 18.18 - 24.04% and T₄ reported higher value. Ether extract varies numerically from 4.26 - 4.62% with T₂ observed to have highest value. An ash content ranges from 07.44 - 11.81% with T₂ having higher value. Nitrogen free extract (NFE) content varies from 50.70 - 53.07% with T₁ recorded highest value. The values obtained in present study reported high dry matter values due to the feed ingredient use in the diets which were all dry materials. The crude protein levels obtained in the present study were lower than the range 16-18% recommended by Norton (2003) for growing sheep and goats but higher than range of 10-12% CP as the moderate level for ruminant production reported by (Gatenby 2002).

Table 2: Proximate Composition of Experimental Diets and Test Material

Parameters	Treatments				PTL*
	TI (0%)	TII (15%)	TIH (30%)	TIV (45%)	
DM	94.30	94.43	94.68	94.88	95.56
CP	13.28	13.63	12.17	13.44	11.61
CF	18.86	18.18	20.97	24.04	25.23
EE	4.26	4.62	4.39	4.38	3.09
Ash	10.53	11.81	10.49	7.44	7.43
NFE	53.07	51.76	51.39	50.70	52.64

PTL* = *Piliostigma thonningii* leaves

Growth Performance

The results of growth performance were presented on Table 3. The values observed in present study revealed final weight, weight gain, average daily gain and feed conversion ratio were significantly ($P<0.05$) higher in T₄ respectively. However, feed intake was significantly ($P<0.05$) higher in T₁ than the values recorded in animals for other treatments. Thus, indicates that the intake reduces as level of *Piliostigma thoningii* leaves increases. The result of growth performance of Kano Brown bucks obtained in the present study could be pointed to the better ability of the goats to tolerate the test diets which accounted for effective feed utilization as observed in feed conversion ratio. The best feed conversion ratio was exhibited by the animals on T₄ (5.60) compared with other treatments, because of lower value obtained, thus the lower the better. However, daily weight gains of goats obtained in this study were higher than the average value of 37.23g/day of goats fed on browse-tree leaves supplement as reported by (Okoruwa and Ikhimioya, 2020). The weight gain recorded in this study corroborate with the earlier reports of Baiden *et al.* (2002) that supplementing diet with protein source will lead to better utilization of the diets by goats. The marked variation in weight gain by the animals fed the experimental diets may be attributed to the inclusion of *Piliostigma thoningii* leaf due to its high nutrient profile and palatability.

Table 3: Effects of graded levels of *Piliostigma thoningii* leaves as complete diets on productive Performance of Kano brown bucks

Parameters (%)	Treatments				LSD
	I (0)	II(15)	III(30)	IV(45)	
IW (kg)	15.50	13.00	14.70	15.25	NS
FW (kg)	18.88 ^d	19.88 ^c	23.88 ^b	25.75 ^a	1.93*
WG (g)	3.38 ^d	6.88 ^c	9.63 ^b	10.5 ^a	0.60*
ADG (g)	43.90 ^d	89.35 ^c	125.06 ^b	136.36 ^a	7.76*
TFC (kg)	67.51	66.60	64.08	62.64	NS
FCR	15.70 ^d	8.29 ^c	5.95 ^b	5.60 ^a	1.11*

^{abc} Means within the same row with different superscript differs significantly ($P<0.05$), IW= initial weight, FW= final weight, WG=weight gain, ADG=Average daily gain, TFC=Total feed consumed, F.I=Feed intake, F.E=Feed efficiency, NS=No significance difference

Nutrient Digestibility

The results of nutrient digestibility of Kano Brown bucks fed complete diet of *Pilostigma thoningii* leaves are shown in Table 4. There were significant ($P<0.05$) differences in all the parameters evaluated. Treatment II had significantly ($P<0.05$) higher value (77.53%) of DMD. The crude protein (CP) digestibility values (86.76) of treatment IV was statistically ($P>0.05$) similar with

treatment III value (86.30) but treatment IV having significantly ($P<0.05$) higher value. For the crude fiber (CF) digestibility, treatment II value (84.63%) was significantly ($P<0.05$) higher than other treatment values. There were significant differences ($P<0.05$) in ether extract (EE) digestibility with higher value recorded in treatment II (83.94%). There were significant ($P<0.05$) difference in ash digestibility among all treatments. NFE digestibility was significantly ($P<0.05$) higher in treatment IV (79.42) and value (78.70) of treatment III was statistically similar. The digestibility of feedstuff is the major determinant of the quality of the feed (Ganiyu *et al.*, 2022). However, digestibility values obtained in this study is in line with report of Okunlola *et al.*, (2023) who stated that the crude protein of the diet which in turn improved the digestibility of the diet. Higher digestibility of the nutrients for T4 may be responsible for the higher weight gained by the animals in these treatments.

Table 4: Nutrient Digestibility (%) of Kano Brown Bucks fed graded levels of *Pilostigma thoningii* as complete diet.

Parameters (%)	Treatments				LSD
	TI (0)	TII (15)	TIII (30)	TIV (45)	
DMD	75.35 ^a	77.53 ^a	70.10 ^b	68.85 ^b	6.85
CPD	80.89 ^b	80.63 ^b	86.30 ^a	86.76 ^a	5.88
CFD	84.40 ^a	84.63 ^a	71.10 ^b	70.91 ^b	8.99
EED	83.88 ^a	83.94 ^a	76.10 ^{ab}	72.79 ^b	10.98
ASHD	75.99 ^a	76.05 ^a	71.22 ^b	70.16 ^b	6.94
NFED	70.13 ^b	70.67 ^b	78.70 ^a	79.42 ^a	6.11

^{abc} Means within the same row with different superscript differs significantly ($P<0.05$), DMD (dry matter digestibility), CPD (crude protein digestibility), CFD (crude fiber digestibility), AshD (ash digestibility), EED (ether extract digestibility) and NFED (nitrogen free extract digestibility), LSD (Least significant difference).

Conclusion

Based on the findings of the present study, it was concluded that *Pilliosigma thonningii* leaves has appreciable crude protein and crude fibre that could be efficiently utilized by ruminant animals. Furthermore, the inclusion of *Pilliosigma thonningii* leaves improved the performance and digestibility of Kano Brown bucks.

Recommendation

It is recommended that 45% inclusion level of *Pilliosigma thonningii* leaves could be incorporated in the diet of Kano brown bucks for optimum performance.

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